

Dissertation Structure — Remaining Chapters (Topline Plan)

This document outlines the high-level structure and purpose of the remaining dissertation chapters following Chapter 3 (System Design). It is intended as a planning scaffold to guide writing and ensure clear separation of concerns across chapters.

Chapter 4 — Implementation

Purpose

To describe how the system design (Chapter 3) was realised in practice. This chapter focuses on **technical execution**, not design justification.

4.1 Implementation Overview

- Brief mapping from design (Chapter 3) to implementation
 - Overview of system components:
 - Unity (orchestration)
 - Bela (input sensing)
 - OSC communication
 - ChuckK (audio playback)
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4.2 Unity System

- Structure of Unity project
 - Key scripts and modules:
 - TurnLoopController
 - Data handling (PatternTurn, buffers)
 - Flow of execution within Unity
 - Interaction between components
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4.3 Bela Integration

- Piezo sensor setup
 - Signal processing pipeline:
 - thresholding
 - peak detection
 - refractory handling
 - Hit event generation
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4.4 OSC Communication Layer

- Message format (`/it4/hit`)
 - Timestamp handling (split 64-bit → reconstruction)
 - Transmission pipeline (Bela → Unity)
 - Reliability considerations
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4.5 Chuck Audio Integration

- Role of Chuck in playback
 - Scheduling events using sample timing
 - Mapping PatternTurn → playback events
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4.6 Debugging and Tooling

- Debug UI components:
 - PatternTurn visualisation
 - planner/skeleton debug panels
 - Runtime observability
 - Development workflow and testing approach
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Key Rule for Chapter 4

- DO: explain *how it is built*
 - DO NOT: re-justify design decisions (already done in Chapter 3)
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Chapter 5 — Evaluation

Purpose

To critically assess the system in terms of **functionality, interaction quality, and limitations**.

5.1 Evaluation Overview

- What is being evaluated
 - Criteria:
 - responsiveness
 - correctness
 - musicality
 - interaction quality
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5.2 Functional Evaluation

- Does the system work as intended?
 - End-to-end pipeline:
 - input → capture → response → playback
 - Stability and reliability
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5.3 Interaction and Musical Behaviour

- Responsiveness to performer input
 - Variation in responses
 - Structural coherence of output
 - Does the system behave like an interaction partner?
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5.4 Response Quality Analysis

- Behaviour of different response types
 - Strengths and weaknesses of generated outputs
 - Observations from testing
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5.5 Limitations

- Flat or predictable responses
 - Lack of tempo adaptation
 - No long-term memory across turns
 - Constraints of rule-based generation
 - Symbolic representation limitations
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5.6 Reflection on Design Decisions

- What worked well
 - What did not work as expected
 - Trade-offs observed in practice
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Key Rule for Chapter 5

- Be **critical and analytical**
 - Avoid purely descriptive writing
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Chapter 6 — Conclusion and Future Work

Purpose

To summarise contributions, reflect on outcomes, and define future directions.

6.1 Summary of Work

- What was built
 - Key system components
 - Research problem revisited
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6.2 Contributions

- Closed-loop interaction architecture
 - Symbolic rhythmic representation
 - Response planning and generation pipeline
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6.3 Limitations (Recap)

- Key system constraints
 - Gaps identified in evaluation
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6.4 Future Work

Potential extensions:

- Adaptive BPM / tempo inference
 - Improved response generation (stochastic / ML-based)
 - Multi-turn memory and context awareness
 - Better use of microtiming
 - Expressive performance modelling
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Key Rule for Chapter 6

- Be concise and reflective
 - Emphasise what has been learned and achieved
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Overall Flow Reminder

- Chapter 3 → Design (WHY and HOW it should work)
 - Chapter 4 → Implementation (HOW it was built)
 - Chapter 5 → Evaluation (HOW WELL it works)
 - Chapter 6 → Conclusion (WHAT it means and WHAT'S NEXT)
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Writing Principle

Each chapter must answer a different question:

- Chapter 4: *How did I build it?*
- Chapter 5: *How well does it work?*
- Chapter 6: *What did I achieve and what next?*

Avoid overlapping these responsibilities.